

Anagha Surendranath

Software Developer & Physicist

EXPERIENCE

April 2024 - October 2025

TutorComp, Kochi, India — Junior Software Engineer

- Developed interactive educational simulations in Next.js covering high school mathematics and physics concepts.
- Engineered custom SVG components to programmatically render dynamic geometric and mathematical visualizations.
- Built modular, responsive components to ensure consistent performance across devices.

November 2020 - October 2022

Aix Marseille University, Marseille, France — Researcher

- Conducted microfluidic experiments to mimic splenic filtration by passing red blood cells (RBCs) through submicron slits.
- Observed that most RBCs could traverse slits as narrow as $0.26\text{ }\mu\text{m}$ at body temperature, significantly smaller than typical RBC dimensions.
- <https://www.pnas.org/doi/10.1073/pnas.2300095120>

EDUCATION

July 2018 - June 2020

Indian Institute of Technology Bombay, Mumbai, India
MSC Physics (CPI 8.86/10)

July 2015 - June 2018

University of Hyderabad, Hyderabad, India
BSC Physics (CGPA 8.82/10)

SELECTED PERSONAL PROJECTS

NoteShare app

Full-stack web application built with React and Node.js (Express.js), implementing CRUD operations for managing application data.

Live Demo - note-share-app-three.vercel.app

Frontend Repository - github.com/anagha-sr/note-share-app

Backend Repository - github.com/anagha-sr/note-share-server

Gaussian Beam Width simulator

React project built with TypeScript, implementing complex physics calculations through an interactive web interface.

Live Demo - gaussian-beam.vercel.app

Repository - https://github.com/anagha-sr/gaussian_beam

 [Windisch, Switzerland](#)

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 github.com/anagha-sr

 [anagha-surendranath-35b1b6191](https://www.linkedin.com/in/anagha-surendranath-35b1b6191)

 portfolio-anagha.vercel.app

ABOUT ME

Software developer with experience building web applications using React and TypeScript. Analytical and adaptable, with strong problem-solving skills and a background in Physics. Quick to learn new technologies and apply them to software development.

CORE SKILLS

- JavaScript
- TypeScript
- React
- HTML
- CSS

EXPOSURE

Node.js · Express.js · MongoDB · Python · Flask · SQLite · Vue.js · Firebase · Java

CERTIFICATIONS

2023 - 2025

Indian Institute of Technology Madras, Chennai, India

Diploma in Programming (CGPA 9.54)

2023 - 2024

Luminar technolab, Kochi, India
MERN Stack course

LANGUAGES

English(Fluent)
Malayalam(Native)
German(A1)